

TCP/IP Communication in Action

Following the data when a user opens a secure website, from the browser to the web server and back

Speakers and responsibilities

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|----|---|--|
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Team Lead, 1st speaker | Introduction + Application layer |
| 02 | Wisdom Binichiuku Odili
2nd speaker | Transport layer + TCP/UDP: how the connection is established |
| 03 | Beverlyn Ewuradwoa Amanfu
3rd speaker | Internet layer, addressing and network devices |
| 04 | Richard Inyang
4th speaker | Encapsulation, decapsulation and the OSI mapping |
| 05 | Fokemg-yalie Brisco Elias
5th speaker | Troubleshooting process |



The TCP/IP model and its four layers

A layered framework that explains how data moves between devices. Each layer has one job.

TCP/IP layer	OSI layers	Main responsibility	Secure website examples
Application	Application, Presentation, Session	Network services directly to the user or application	DNS, HTTP/HTTPS, browser
Transport	Transport	End-to-end communication between applications	TCP, UDP; TCP port 443
Internet	Network	Logical addressing and routing between networks	IP, ICMP
Network Access	Data Link, Physical	Moves frames across the local network and physical medium	Ethernet/Wi-Fi, MAC addresses, switches

TCP/IP has four layers; the OSI model has seven.

Scenario: opening a secure website

A user types `https://www.icdfa.edu.ng` in the browser and presses Enter.

01

Enter URL

User types the address in the browser

02

Read URL

Browser interprets the address

03

DNS lookup

Domain name becomes an IP address

04

TCP connect

Three-way handshake to port 443

05

HTTPS request

TLS-protected HTTP request is sent

06

Response

Server replies along the reverse path

Data is encapsulated as it moves down the layers on the client, and decapsulated as it moves up the layers at the web server.

Application layer: how the communication starts

DNS: finding the IP address

- 1 Client checks its local DNS cache
- 2 If needed, a query goes to a DNS resolver
- 3 Resolver returns the IP address
- 4 Browser uses it to contact the web server

Like phone contacts: select a name and the phone finds the number. DNS does this for website names.

HTTPS: HTTP protected by TLS

HTTP carries web content between browser and server. TLS protects the channel.

Confidentiality

Helps stop others simply reading the traffic

Integrity

Helps detect unauthorised changes to data

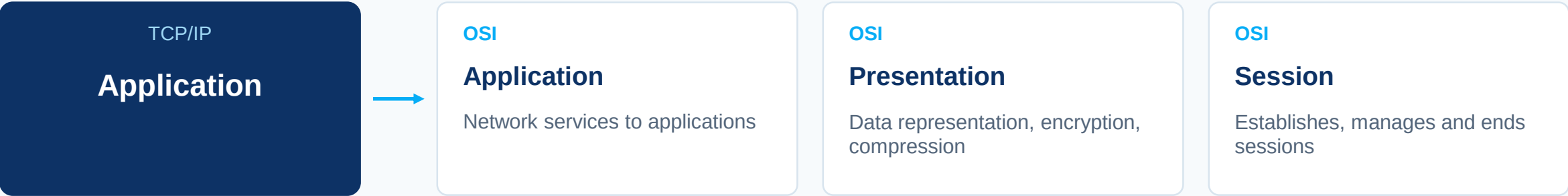
Authentication

Helps the browser verify the server's identity

DNS, HTTP and HTTPS belong to the TCP/IP Application layer.

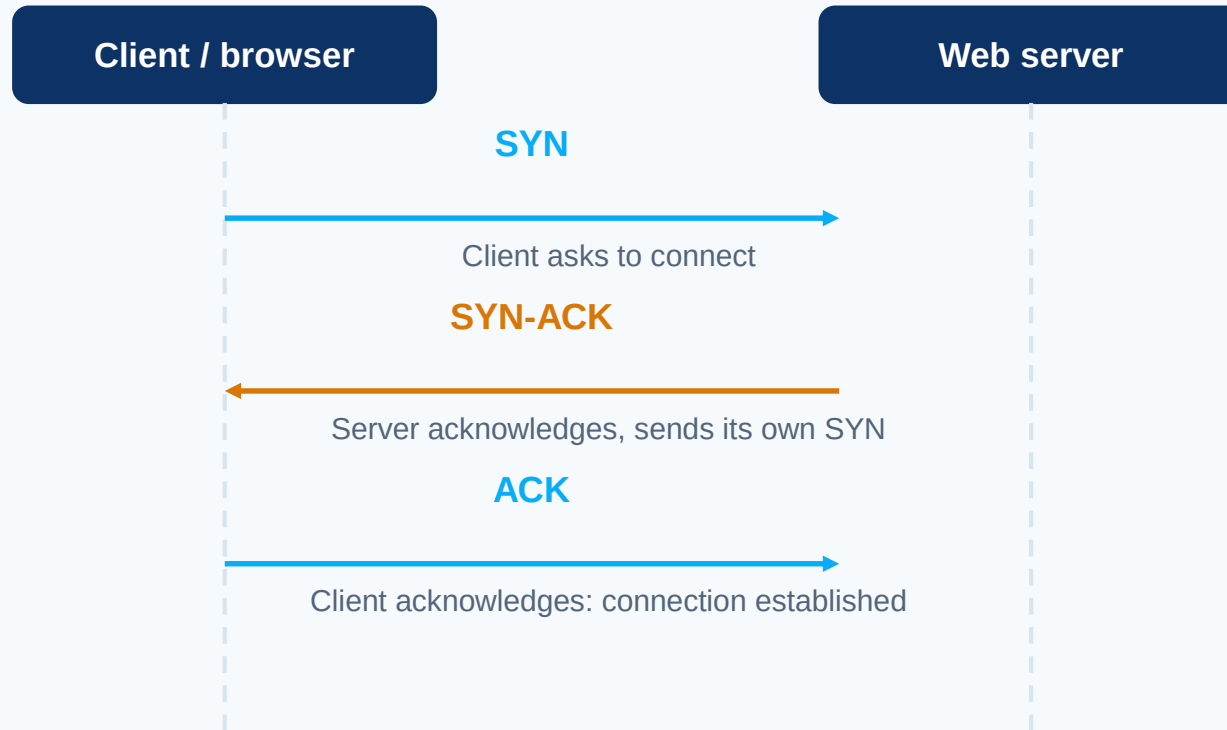
Application layer: protocols and the OSI model

Protocol	Meaning	Purpose
DNS	Domain Name System	Finds the IP address of a domain
HTTP	Hypertext Transfer Protocol	Transfers web content
HTTPS	Hypertext Transfer Protocol Secure	Protects web communication using TLS



TLS functions sit near OSI Presentation/Session, though modern stacks do not map one-to-one.

Transport layer: the TCP three-way handshake



After the handshake, HTTPS/TLS communication can begin.

Destination port: TCP 443

Identifies the HTTPS service on the server

Source port: ephemeral

A temporary port chosen by the client's operating system

Reliability

Sequence numbers, acknowledgements and retransmission keep data complete and in order

TCP compared with UDP

Feature	TCP	UDP
Connection	Connection-oriented; uses a handshake	Connectionless; no TCP-style handshake
Reliability	Acknowledgements, retransmission and ordered delivery	No TCP-style guaranteed delivery or ordering
Speed / overhead	More overhead from connection and reliability mechanisms	Lower overhead; suits small or time-sensitive traffic
Example uses	Traditional HTTPS over TCP, file transfer, email protocols	DNS queries, streaming, real-time apps, HTTP/3 via QUIC

UDP is not simply a faster TCP. It is a different service model.

Internet layer: addressing, routing and devices

IP addressing

IPv4

Four decimal numbers, e.g. 192.168.1.50

IPv6

Hexadecimal with colons, e.g. 2002:db8::1

Private / public

Private inside local networks; public is routable online

Source → destination

Sender's IP to the web server's IP, found through DNS

Network devices

Router

Reads the destination IP and forwards packets between networks

Switch

Connects devices inside a local network

Wi-Fi access point

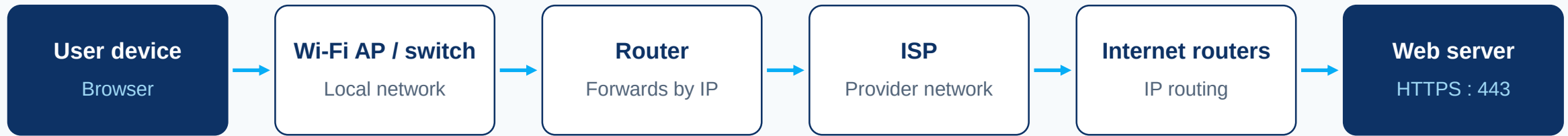
Gives wireless access to the local network

ISP

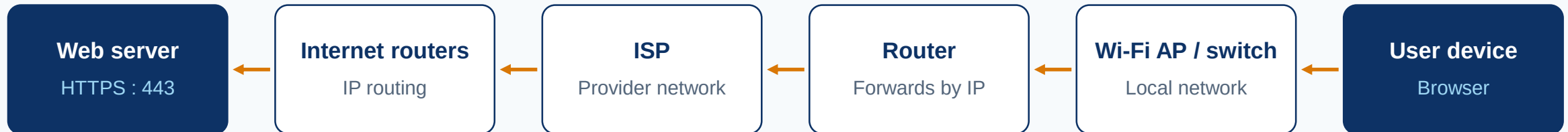
Carries traffic from the local network to the wider Internet

Packet journey: client to web server

REQUEST



RESPONSE



The router reads the destination IP and forwards the packet. The response follows the same path back and is decapsulated at the user's device.

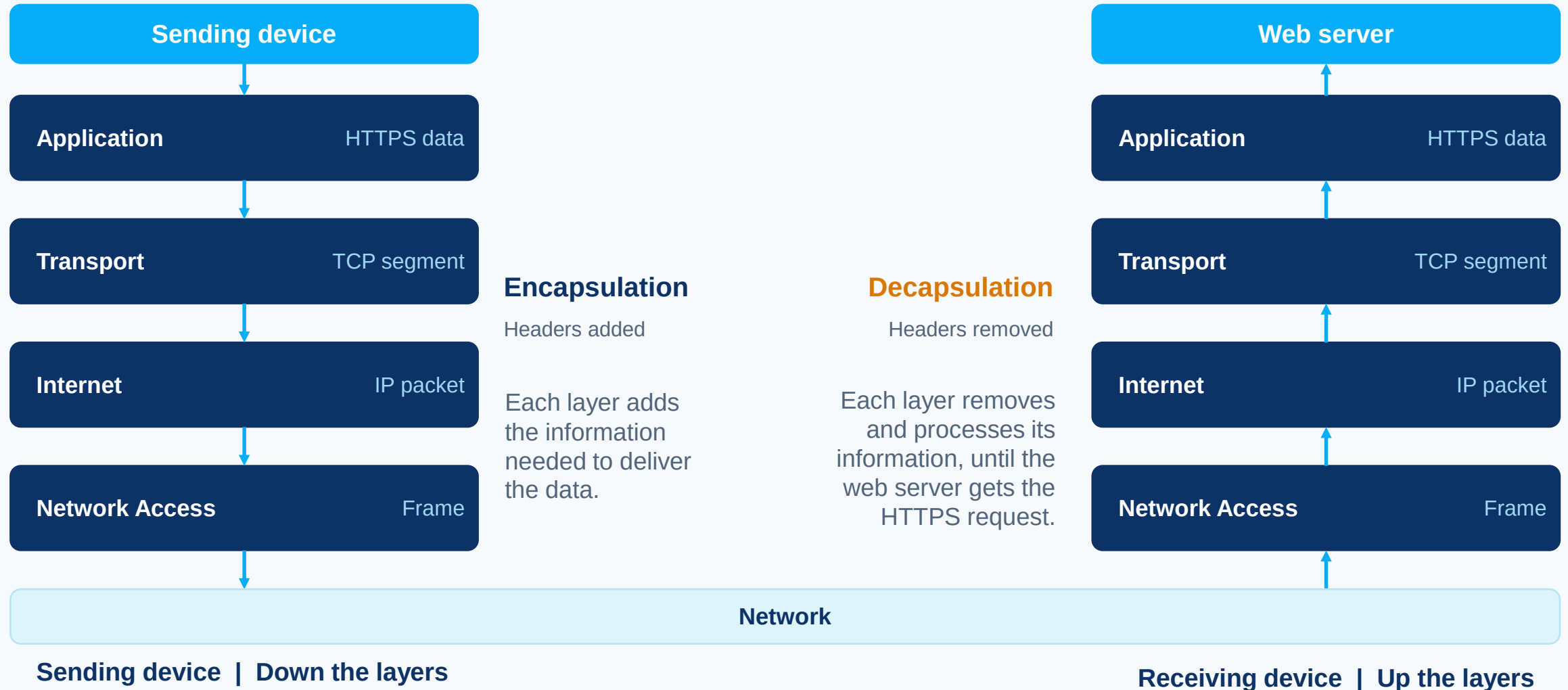
The packet may pass through several routers before reaching the web server.

IP address vs MAC address

Feature	IP address	MAC address
Other name	Logical address	Physical address
Layer	Internet layer	Network Access layer
Purpose	Identifies the network location of a host	Identifies a network interface on the local link
Example	192.168.1.50	A1:B4:C5:DD:C1:3E
Used for	Routing between networks	Local network / frame delivery

IP address = where the device is on the network. MAC address = which interface is addressed on the local link.

Complete view: encapsulation and decapsulation



Troubleshooting: the website does not load

1

Check IP configuration

ipconfig or ip addr

2

Test the default gateway

ping gateway IP

3

Test external connectivity

ping 8.8.8.8

4

Check DNS resolution

nslookup or dig

5

Check destination and routing

Route and router reachability

6

Check TCP port 443

Blocked, refused or dropped?

7

Check firewall rules

Client, network, router, server

8

Capture packets

Wireshark, if still unclear

9

Interpret the TCP handshake

SYN, SYN-ACK, ACK or RST

10

Test the website again

Confirm the fix

Reading the handshake

Repeated SYNs, no SYN-ACK

Possible connectivity, firewall, routing or server issue. Needs further investigation.

Immediate RST

The destination may be refusing the connection.

SYN, SYN-ACK, ACK

Handshake completes, HTTPS/TLS starts and the page loads.

Reference

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Thank you.